

Hard Drives

IDE Interface

Abbreviation of either Intelligent Drive Electronics or Integrated Drive Electronics. The IDE interface is an interface for mass-storage devices, in which the controller is integrated into the disk or CD-ROM drive. Although it really refers to a general technology, most people use the term to refer to the ATA specification which uses this technology.

EIDE

Short for Enhanced IDE, a newer version of the IDE device interface standard developed by Western Digital Corporation. Supports data rates of between 4 and 16.6 MBps, about three to four times faster than the older IDE standard. In addition, it can support mass-storage devices of up to 8.4 gigabytes, whereas the old standard was limited to 528 MB. EIDE is sometimes referred to as Fast ATA or Fast IDE, which is essentially the same standard, developed by Seagate. It is also sometimes called ATA-2.

There are four EIDE modes defined. The most common is Mode 4, which supports transfer rates of 16.6 MBps. There is also a mode called ATA-3 or Ultra ATA, that supports transfer rates of 33 MBps.

ATA

A drive implementation that integrates the controller on the drive itself. There are several versions of ATA, all developed by the Small Form Factor (SFF) Committee:

ATA: Known also as IDE, supports one or two hard drives, a 16-bit interface and PIO modes 0, 1 and 2.

ATA-2: Supports faster PIO modes (3 and 4) and multiword DMA modes (1 and 2). Also supports logical block addressing (LBA) and block transfers. ATA-2 is marketed as Fast ATA and Enhanced IDE (EIDE). ATA-2 has an extension called ATAPI.